

Desafio 2

Question 1.

Podemos escrever $\mathbf{X}'\mathbf{A}\mathbf{X}$ em termos dos desvios em relação à média:

$$\begin{aligned} \mathbf{X}'\mathbf{A}\mathbf{X} &= (\mathbf{X} - \boldsymbol{\theta} + \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta} + \boldsymbol{\theta}) = [(\mathbf{X} - \boldsymbol{\theta}) + \boldsymbol{\theta}]'\mathbf{A}[(\mathbf{X} - \boldsymbol{\theta}) + \boldsymbol{\theta}] \\ (1) \quad &= \underbrace{(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})}_{1^\circ \text{ termo}} + \underbrace{(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}\boldsymbol{\theta}}_{2^\circ \text{ termo}} + \underbrace{\boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})}_{3^\circ \text{ termo}} + \underbrace{\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}}_{4^\circ \text{ termo}} \end{aligned}$$

No segundo termo, $(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}\boldsymbol{\theta}$ é um escalar:

$$\begin{aligned} (\mathbf{X} - \boldsymbol{\theta}) &= \begin{bmatrix} x_1 - \theta_1 \\ x_2 - \theta_2 \\ \vdots \\ x_m - \theta_m \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mm} \end{bmatrix}, \quad \boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_m \end{bmatrix} \\ (\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}\boldsymbol{\theta} &= \sum_{k=1}^m (x_k - \theta_k) \sum_{l=1}^m a_{kl} \theta_l = \sum_{k=1}^m \sum_{l=1}^m (x_k - \theta_k) a_{kl} \theta_l \end{aligned}$$

Um escalar é igual à sua transposta, portanto:

$$[(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}\boldsymbol{\theta}]' = \boldsymbol{\theta}'\mathbf{A}'(\mathbf{X} - \boldsymbol{\theta})$$

Como $\mathbf{A}$ é simétrica, temos que $\mathbf{A} = \mathbf{A}'$. Então:

$$\boldsymbol{\theta}'\mathbf{A}'(\mathbf{X} - \boldsymbol{\theta}) = \boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta}) \Rightarrow (\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}\boldsymbol{\theta} = \boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})$$

Portanto, o 2º e 3º termos são iguais, e assim, reescrevendo (1):

$$\begin{aligned} \mathbf{X}'\mathbf{A}\mathbf{X} &= (\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta}) + \boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta}) + (\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}\boldsymbol{\theta} + \boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta} \\ &= (\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta}) + 2\boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta}) + \boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta} \end{aligned}$$

Cálculo da variância de $\mathbf{X}'\mathbf{A}\mathbf{X}$:

$$\text{Var}(\mathbf{X}'\mathbf{A}\mathbf{X}) = \mathbb{E}[(\mathbf{X}'\mathbf{A}\mathbf{X})^2] - [\mathbb{E}(\mathbf{X}'\mathbf{A}\mathbf{X})]^2$$

1. Calculando $\mathbb{E}[(\mathbf{X}'\mathbf{A}\mathbf{X})]^2$

Temos que:

$$\begin{aligned} \mathbb{E}[\mathbf{X}'\mathbf{A}\mathbf{X}] &= \text{tr}(\mathbb{E}[\mathbf{X}'\mathbf{A}\mathbf{X}]) \\ &= \mathbb{E}[\text{tr}(\mathbf{X}'\mathbf{A}\mathbf{X})] \\ &= \mathbb{E}[\text{tr}(\mathbf{A}\mathbf{X}\mathbf{X}')] \\ &= \text{tr}(\mathbb{E}[\mathbf{A}\mathbf{X}\mathbf{X}']) \\ &= \text{tr}(\mathbf{A}\mathbb{E}[\mathbf{X}\mathbf{X}']) \\ &= \text{tr}(\mathbf{A}(\text{Var}[\mathbf{X}] + \boldsymbol{\theta}\boldsymbol{\theta}')) \\ &= \text{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \text{tr}(\mathbf{A}\boldsymbol{\theta}\boldsymbol{\theta}') \\ &= \text{tr}(\mathbf{A}\mu_2\mathbf{I}) + \boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta} \\ &= \mu_2 \text{tr}(\mathbf{A}) + \boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta} \end{aligned}$$

Logo,

$$\begin{aligned}\mathbb{E}[(\mathbf{X}'\mathbf{A}\mathbf{X})]^2 &= (\boldsymbol{\mu}_2 \text{tr}(\mathbf{A}) + \boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2 \\ &= \boldsymbol{\mu}_2^2 \text{tr}(\mathbf{A})^2 + 2\boldsymbol{\mu}_2 \text{tr}(\mathbf{A})\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta} + (\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2\end{aligned}$$

2. Calculando $\mathbb{E}[(\mathbf{X}'\mathbf{A}\mathbf{X})^2]$

Temos que:

$$\begin{aligned}(\mathbf{X}'\mathbf{A}\mathbf{X})^2 &= [(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})]^2 + 4[\boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})]^2 + (\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2 \\ &\quad + 2\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}[(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta}) + 4\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}\boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})] \\ &\quad + 4\boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})\end{aligned}$$

Logo,

$$\begin{aligned}\mathbb{E}(\mathbf{X}'\mathbf{A}\mathbf{X})^2 &= \mathbb{E}[(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})]^2 + \mathbb{E}[4(\boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta}))^2] \\ &\quad + \mathbb{E}[(\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2] + \mathbb{E}[2\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta} \cdot (\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})] \\ &\quad + \mathbb{E}[4\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta} \cdot \boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})] \\ &\quad + \mathbb{E}[4\boldsymbol{\theta}'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})(\mathbf{X} - \boldsymbol{\theta})'\mathbf{A}(\mathbf{X} - \boldsymbol{\theta})]\end{aligned}$$

Assumindo $\mathbf{Y} = \mathbf{X} - \boldsymbol{\theta}$, e consequentemente, $\mathbb{E}[\mathbf{Y}] = \mathbf{0}$, temos que:

$$\begin{aligned}\mathbb{E}[(\mathbf{X}'\mathbf{A}\mathbf{X})^2] &= \mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2] + 4\mathbb{E}[(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2] + (\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2 \\ &\quad + 2\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}(\mathbb{E}[\mathbf{Y}'\mathbf{A}\mathbf{Y}] + 4\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta} \cdot \boldsymbol{\theta}'\mathbf{A}\mathbb{E}[\mathbf{Y}]) \\ &\quad + 4\mathbb{E}[\boldsymbol{\theta}'\mathbf{A}\mathbf{Y}\mathbf{Y}'\mathbf{A}\mathbf{Y}]\end{aligned}$$

Temos que, $\mathbb{E}[\mathbf{Y}'\mathbf{A}\mathbf{Y}] = \boldsymbol{\mu}_2 \text{tr}(\mathbf{A})$ e $\mathbb{E}[\mathbf{Y}] = \mathbf{0}$. Então

$$\mathbb{E}[(\mathbf{X}'\mathbf{A}\mathbf{X})^2] = \mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2] + 4\mathbb{E}[(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2] + (\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta})^2 + 2\boldsymbol{\theta}'\mathbf{A}\boldsymbol{\theta}\boldsymbol{\mu}_2 \text{tr}(\mathbf{A}) + 4\mathbb{E}[\boldsymbol{\theta}'\mathbf{A}\mathbf{Y}\mathbf{Y}'\mathbf{A}\mathbf{Y}]$$

Vamos calcular cada termo da expressão acima:

- $\mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2]$:

$(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2$ pode ser escrito da seguinte maneira:

$$(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2 = \left(\sum_{i,j} a_{ij} Y_i Y_j \right)^2 = \sum_{i,j,k,l} a_{ij} a_{kl} Y_i Y_j Y_k Y_l = \sum_i \sum_j \sum_k \sum_l a_{ij} a_{kl} Y_i Y_j Y_k Y_l$$

$$\mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2] = \sum_{i,j,k,l} a_{ij} a_{kl} \mathbb{E}(Y_i Y_j Y_k Y_l)$$

Agora, precisamos calcular $\mathbb{E}(Y_i Y_j Y_k Y_l)$ usando as hipóteses de independência e momentos centrados:

$$\mathbb{E}[Y_i Y_j Y_k Y_l] = \begin{cases} \mu_4, & \text{se } i = j = k = l, \\ \mu_2^2, & \text{se formam dois pares iguais:} \\ & (i = j \text{ e } k = l) \text{ ou } (i = k \text{ e } j = l) \text{ ou } (i = l \text{ e } j = k), \\ 0, & \text{caso contrário.} \end{cases}$$

Dessa forma:

1) **Caso** $i = j = k = l$:

$$\mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2] = \sum_i a_{ii}^2 \mu_4$$

2) **Caso** $i = j, k = l$ (com $k \neq i$):

$$\mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2] = \sum_i \sum_{k \neq i} a_{ii} a_{kk} \mu_2^2$$

3) **Caso** $i = k, j = l$ (**com** $j \neq i$):

$$\mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2] = \sum_i \sum_{j \neq i} a_{ij}^2 \mu_2^2$$

4) **Caso** $i = l, j = k$ (**com** $j \neq i$):

$$\mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2] = \sum_i \sum_{j \neq i} a_{ij} a_{ji} \mu_2^2$$

Portanto, somando todos os casos temos que:

$$\begin{aligned} \mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2] &= \mu_4 \sum_i a_{ii}^2 + \mu_2^2 \left[\sum_i \sum_{k \neq i} a_{ii} a_{kk} \right. \\ &\quad \left. + \sum_i \sum_{j \neq i} a_{ij}^2 + \sum_i \sum_{j \neq i} a_{ij} a_{ji} \right] \end{aligned}$$

Note, que como $\mathbf{A}$ é simétrica, vale que $a_{ij} = a_{ji}$, então:

$$\sum_i \sum_{j \neq i} a_{ij} a_{ji} = \sum_i \sum_{j \neq i} a_{ij}^2 = \text{tr}(\mathbf{A}^2)$$

Temos também que:

$$\sum_i \sum_{k \neq i} a_{ii} a_{kk} = \sum_i a_{ii} \sum_{k \neq i} a_{kk} = \text{tr}(\mathbf{A}) \cdot \text{tr}(\mathbf{A}) = [\text{tr}(\mathbf{A})]^2$$

Além disso, a expressão $\mu_4 \sum_i a_{ii}^2$ pode ser reescrita como $(\mu_4 - 3\mu_2^2) \mathbf{a}'\mathbf{a}$.

Dessa forma,

$$\mathbb{E}[(\mathbf{Y}'\mathbf{A}\mathbf{Y})^2] = (\mu_4 - 3\mu_2^2) \mathbf{a}'\mathbf{a} + \mu_2^2 [\text{tr}(\mathbf{A})^2 + 2 \text{tr}(\mathbf{A}^2)]$$

- $4\mathbb{E}[(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2]$:

Considerando $\mathbf{b} = \mathbf{A}\boldsymbol{\theta}$, teremos que:

$$(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2 = (\mathbf{A}\boldsymbol{\theta})'\mathbf{Y})^2 = (\mathbf{b}'\mathbf{Y})^2 = \left(\sum_i \mathbf{b}_i \mathbf{Y}_i \right)^2 = \sum_i \sum_j \mathbf{b}_i \mathbf{b}_j \mathbf{Y}_i \mathbf{Y}_j$$

Dessa forma,

$$\mathbb{E}[(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2] = \sum_i \sum_j \mathbf{b}_i \mathbf{b}_j \mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j)$$

Com relação a $\mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j)$:

Sabendo que $\mu_r = \mathbb{E}[(X_i - \theta_i)^r]$, logo temos para $\mathbf{Y}$ que

$$\mu_r = \mathbb{E}[(\mathbf{Y}_i + \boldsymbol{\theta}_i - \boldsymbol{\theta}_i)^r] = \mathbb{E}[(\mathbf{Y}_i)^r]$$

- Se $i = j$, então $\mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j) = \mathbb{E}(\mathbf{Y}_i^2) = \mu_2$
- Se $i \neq j$, então $\mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j) = 0$

Assim, somando todos os casos:

$$4 \mathbb{E}[(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y})^2] = 4 \sum_i \mathbf{b}_i^2 \mu_2 + 0 = 4\mu_2 \sum_i \mathbf{b}_i^2 = 4\mu_2 \mathbf{b}'\mathbf{b} = 4\mu_2 (\mathbf{A}\boldsymbol{\theta})'(\mathbf{A}\boldsymbol{\theta}) = 4\mu_2 \boldsymbol{\theta}'\mathbf{A}'\mathbf{A}\boldsymbol{\theta} = 4\mu_2 \boldsymbol{\theta}'\mathbf{A}^2\boldsymbol{\theta}$$

- $4\mathbb{E}[\boldsymbol{\theta}'\mathbf{A}\mathbf{Y}\mathbf{Y}'\mathbf{A}\mathbf{Y}]$:

Assumindo $\mathbf{b} = \mathbf{A}\boldsymbol{\theta}$, temos que:

$$\boldsymbol{\theta}'\mathbf{A}\mathbf{Y}\mathbf{Y}'\mathbf{A}\mathbf{Y} = \sum_i \sum_j \sum_k \mathbf{b}_i \mathbf{a}_{jk} \mathbf{Y}_i \mathbf{Y}_j \mathbf{Y}_k$$

Logo,

$$\mathbb{E}(\boldsymbol{\theta}'\mathbf{A}\mathbf{Y}\mathbf{Y}'\mathbf{A}\mathbf{Y}) = \sum_i \sum_j \sum_k \mathbf{b}_i \mathbf{a}_{jk} \mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j \mathbf{Y}_k)$$

Se $i = j = k$, então $\mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j \mathbf{Y}_k) = \mathbb{E}(\mathbf{Y}_i^3) = \mu_3$

Se entre i, j e k houver pelo menos um par diferente, então $\mathbb{E}(\mathbf{Y}_i \mathbf{Y}_j \mathbf{Y}_k) = 0$

Assim,

$$4 \mathbb{E}[\boldsymbol{\theta}' \mathbf{A} \mathbf{Y} \mathbf{Y}' \mathbf{A} \mathbf{Y}] = 4 \sum_i \mathbf{b}_i \mathbf{a}_{ii} \mu_3 = 4 \mu_3 \sum_i \mathbf{b}_i \mathbf{a}_{ii} = 4 \mu_3 \mathbf{b}' \mathbf{a} = 4 \mu_3 \boldsymbol{\theta}' \mathbf{A} \mathbf{a}$$

Portanto,

$$\begin{aligned} \mathbb{E}[(\mathbf{X}' \mathbf{A} \mathbf{X})^2] &= \mathbb{E}[(\mathbf{Y}' \mathbf{A} \mathbf{Y})^2] + 4 \mathbb{E}[(\boldsymbol{\theta}' \mathbf{A} \mathbf{Y})^2] + (\boldsymbol{\theta}' \mathbf{A} \boldsymbol{\theta})^2 \\ &\quad + 2 \boldsymbol{\theta}' \mathbf{A} \boldsymbol{\theta} \mathbb{E}(\mathbf{Y}' \mathbf{A} \mathbf{Y}) + 4 \mu_3 \boldsymbol{\theta}' \mathbf{A} \mathbf{a} \\ &= (\mu_4 - 3 \mu_2^2) \mathbf{a}' \mathbf{a} + \mu_2^2 [\text{tr}(\mathbf{A})^2 + 2 \text{tr}(\mathbf{A}^2)] \\ &\quad + 4 \mu_2 \boldsymbol{\theta}' \mathbf{A}^2 \boldsymbol{\theta} + (\boldsymbol{\theta}' \mathbf{A} \boldsymbol{\theta})^2 + 2 \mu_2 \boldsymbol{\theta}' \mathbf{A} \boldsymbol{\theta} \text{tr}(\mathbf{A}) + 4 \mu_3 \boldsymbol{\theta}' \mathbf{A} \mathbf{a} \end{aligned}$$

Por fim, voltando à fórmula da variância, substituindo os valores, temos:

$$\begin{aligned} \text{Var}(X'AX) &= \mathbb{E}[(\mathbf{X}' \mathbf{A} \mathbf{X})^2] - [\mathbb{E}(\mathbf{X}' \mathbf{A} \mathbf{X})]^2 \\ &= \left[(\mu_4 - 3 \mu_2^2) \mathbf{a}' \mathbf{a} + \mu_2^2 (\text{tr}(\mathbf{A})^2 + 2 \text{tr}(\mathbf{A}^2)) + 4 \mu_2 \boldsymbol{\theta}' \mathbf{A}^2 \boldsymbol{\theta} + (\boldsymbol{\theta}' \mathbf{A} \boldsymbol{\theta})^2 \right. \\ &\quad \left. + 2 \mu_2 \boldsymbol{\theta}' \mathbf{A} \boldsymbol{\theta} \text{tr}(\mathbf{A}) + 4 \mu_3 \boldsymbol{\theta}' \mathbf{A} \mathbf{a} \right] \\ &\quad - \left[\mu_2^2 \text{tr}(\mathbf{A})^2 + 2 \mu_2 \text{tr}(\mathbf{A}) \boldsymbol{\theta}' \mathbf{A} \boldsymbol{\theta} + (\boldsymbol{\theta}' \mathbf{A} \boldsymbol{\theta})^2 \right] \\ &= (\mu_4 - 3 \mu_2^2) \mathbf{a}' \mathbf{a} + 2 \mu_2^2 \text{tr}(\mathbf{A}^2) + 4 \mu_2 \boldsymbol{\theta}' \mathbf{A}^2 \boldsymbol{\theta} + 4 \mu_3 \boldsymbol{\theta}' \mathbf{A} \mathbf{a} \end{aligned}$$

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